



Parramatta High School

2022

TRIAL EXAMINATION

Physics

General

- Reading time – 5 minutes
- Working time – 3 hours

Instructions

- Write using black pen
- Draw diagrams using pencil
- NESA approved calculators may be used
- A formula sheet and data sheet and Periodic Table are provided at the back of this paper.
- For questions in Section II, **show all relevant working** in questions involving calculations

Total marks 100

Section I -20 marks (pages 2-9)

- Attempt Questions 1– 20
- Allow about 35 minutes for this part

Section II -80 marks (pages 11-24)

- Attempt Questions 21–34
- Allow about 2 hour and 25 minutes for this part

This paper MUST NOT be removed from the examination room

Section I

20 marks

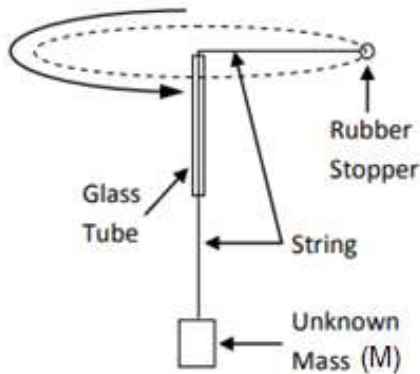
Attempt questions 1 – 20

Allow about 35 minutes for this part

Use the multiple-choice answer sheet for Questions 1-20.

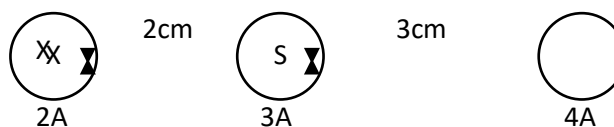
- Which of the following is one piece of information that Maxwell used to formulate his theory on electromagnetism?
(A) Magnetic field lines always form closed loops.
(B) Electric fields are generated by moving charges.
(C) A varying electric field induces a magnetic field.
(D) Electric field lines originate on charges and terminate on positive charges.
- Which the following is a similarity between a motor and a generator?
(A) They both rely on Faraday's Law.
(B) They both require an input current to work.
(C) They both can be explained by Ampere's Law.
(D) They both need to be rotated by an external force.
- What is the angle between the direction of the velocity and acceleration at the highest point of a projectile path?
(A) Zero
(B) 45°
(C) 90°
(D) 180°
- Which of the following did Millikan apply in his oil drop experiment calculations?
(A) Newton's Third Law
(B) Newton's Second law
(C) Law of conservation of charge
(D) The second law of thermodynamics
- What conclusion can be made based on the result of the cathode ray tube experiment with the paddle wheel?
(A) Cathode rays are particles because they interact with the gravitational field.
(B) Cathode rays are particles because they have momentum causing the wheel to spin.
(C) Cathode rays are waves because they interact with the electric field and magnetic field.
(D) Cathode rays are waves because they transfer their energy to the wheel, causing it to spin.

- The diagram below shows a rubber stopper with constant mass m undergoing uniform circular motion at a velocity v with constant radius r . A hanging mass of M is used to counteract the outward pull of the rubber stopper.



What is the expression used to find the value of the unknown hanging mass?

- (A) $\frac{mv}{gr}$
 - (B) $\frac{mv^2}{r}$
 - (C) $\frac{mv^2}{gr}$
 - (D) $\frac{mv^2}{g}$
- Three 20 cm long parallel wires carry currents of 2A, 3A and 4A, as shown below.



What is the net force acting on the middle wire?

- (A) 2.8×10^{-5} N to the left
 - (B) 0.4×10^{-5} N to the left
 - (C) 0.4×10^{-5} N to the right
 - (D) 2.8×10^{-5} N to the right
- A satellite of mass 80 kg moves from a point where the gravitational potential due to the Earth is -15 MJ kg^{-1} to another point where the gravitational potential is -65 MJ kg^{-1} .
- In which direction does the satellite move, and what is its change in potential energy?
- (A) Closer to the Earth and a loss of 50 MJ of potential energy
 - (B) Closer to the Earth and a loss of 4000 MJ of potential energy
 - (C) Further from the Earth and a gain of 50 MJ of potential energy
 - (D) Further from the Earth and a gain of 4000 MJ of potential energy

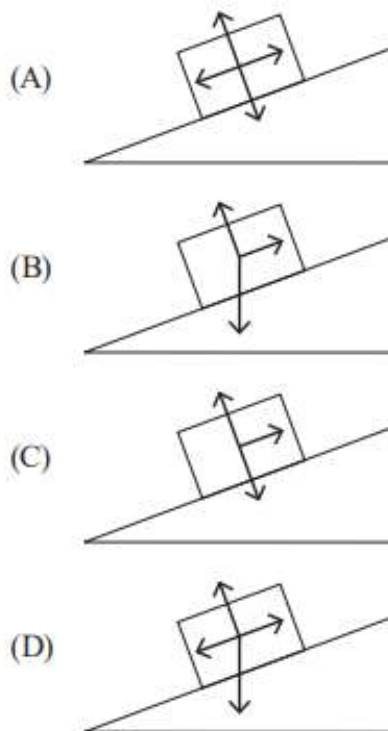
- Which of the following is a true statement about historical attempts to understand the nature of light?
 - (A) The double-slit experiment, the success of Huygen's principle, and the photoelectric effect all favoured the wave interpretation of light.
 - (B) The double-slit experiment, the success of Huygen's principle, and the photoelectric effect all favoured the particle interpretation of light.
 - (C) The double-slit experiment and the success of Huygen's principle favoured the wave interpretation of light, and the photoelectric effect favoured the particle interpretation.
 - (D) The success of Huygen's principle favoured the wave interpretation of light, and the double-slit experiment and the photoelectric effect favoured the particle interpretation.

- A charged particle moves through a region of space at a constant speed, without deflecting (ignoring gravitational field).

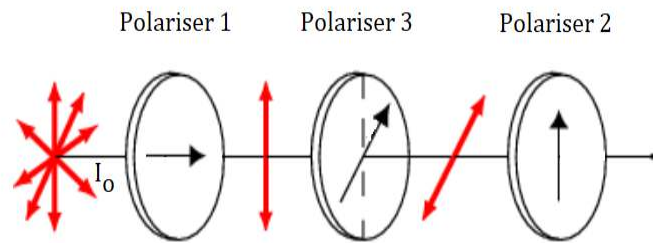
Which of the following statement is correct?

- (A) It cannot be determined since it depends on the charge of the particle.
- (B) There could be electric and magnetic fields, oriented in the same direction.
- (C) There could be electric and magnetic fields, oriented in the opposite direction.
- (D) There could be electric and magnetic fields, oriented perpendicular to each other.

- Which diagrams correctly represent the forces acting on an object resting on an inclined plane?



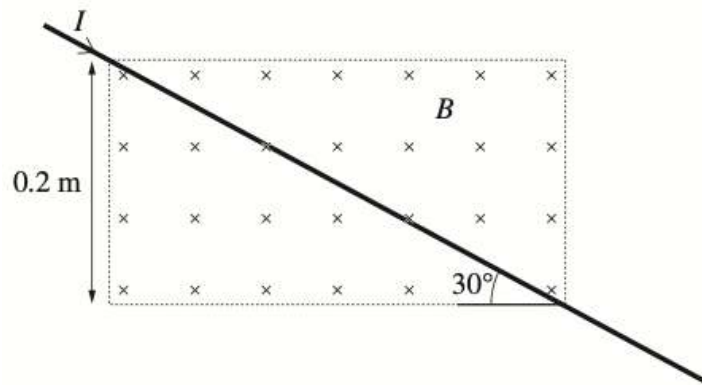
- A source of unpolarised light is initially viewed through two ideal polarisers, one with a transmission axis in the x-direction and one with a transmission axis in the y-direction. Later a third ideal polariser with a transmission axis at 60° from the x-direction is placed between the first two polarisers, as shown below.



Which of the following statements is true?

- (A) Initially, $\frac{1}{2}I_0$ light passes through the first two polarisers, and $\frac{1}{4}I_0$ light passes through after the third polariser is inserted.
- (B) Initially, no light passes through the first two polarisers, and $\frac{1}{4}I_0$ light passes through after the third polariser is inserted.
- (C) Initially, no light passes through the first two polarisers, and $\frac{3}{8}I_0$ light passes through after the third polariser is inserted.
- (D) Initially, no light passes through the first two polarisers, and $\frac{3}{4}I_0$ light passes through after the third polariser is inserted.

- A current-carrying wire passes through a region of a uniform magnetic field, as shown below.

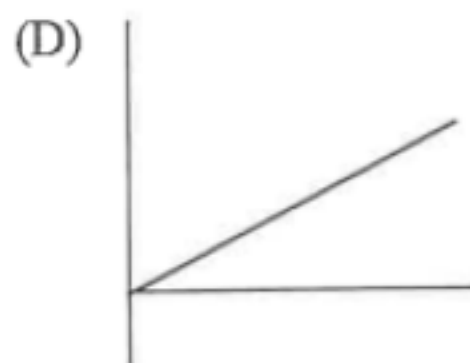
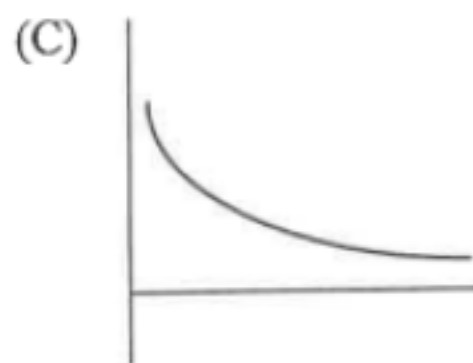
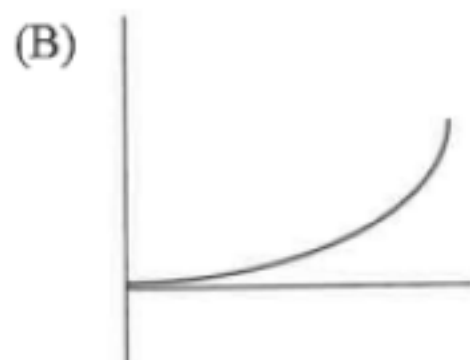
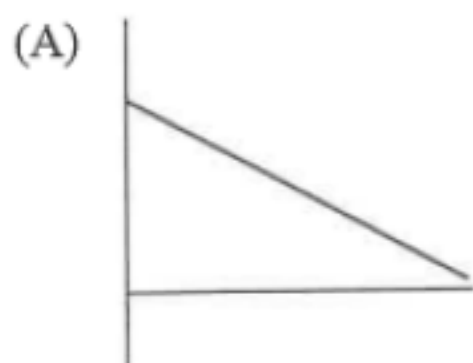


If the current is 1.7A and the wire experiences a force of magnitude 0.034N.

What is the magnitude of the magnetic field?

- (A) 0.05T
- (B) 0.06T
- (C) 0.10T
- (D) 0.20T

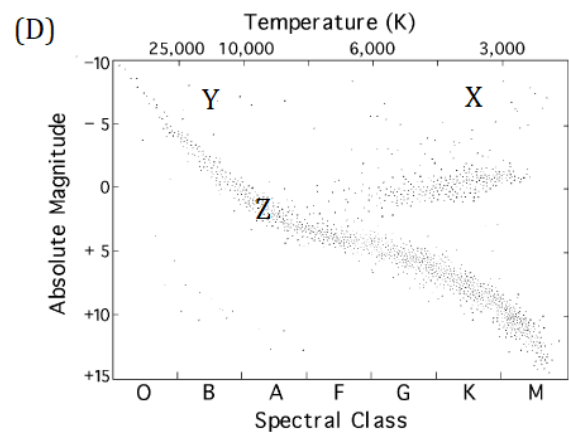
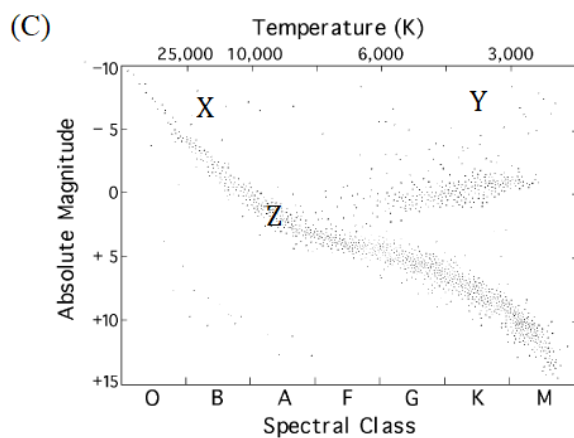
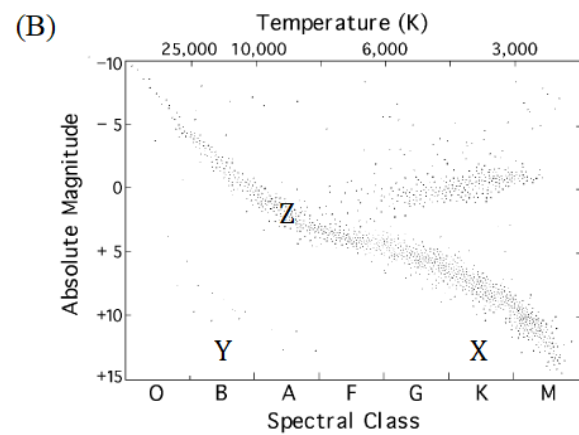
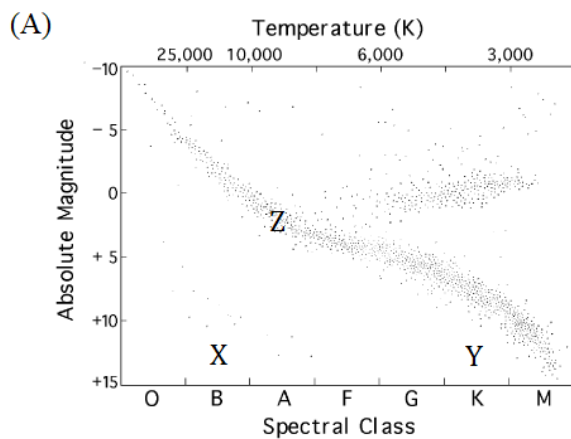
- Which graph shows the relationship between the mass and gravitational field of planets with the same radius?



- The table shows the luminosity and spectra lines for Star X, Y, and Z.

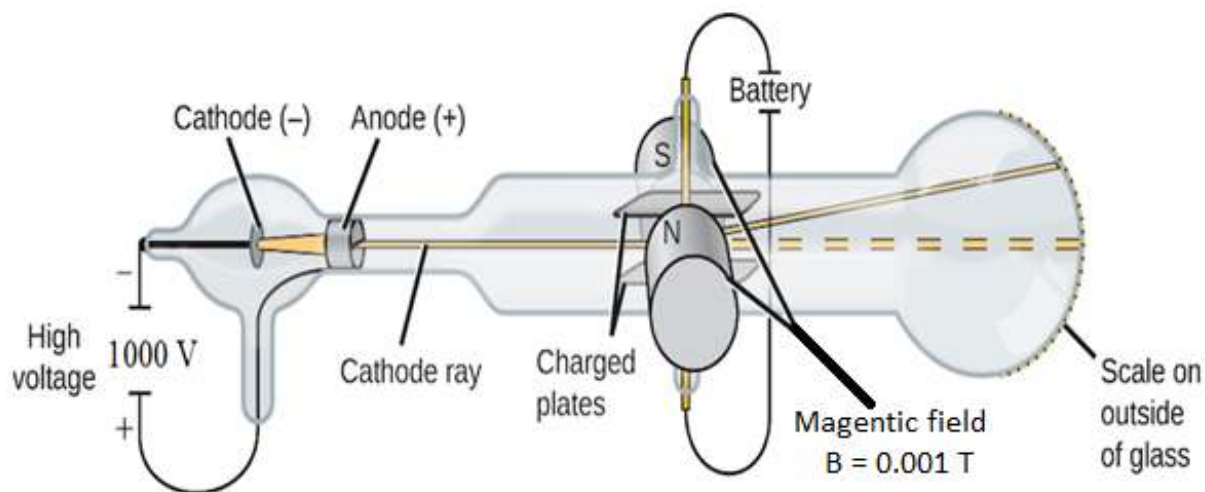
Star	Luminosity ($Sun=1$)	Spectra lines
X	0.0001	Helium and some hydrogen
Y	0.0001	Neutral metals, sodium
Z	30	Strong hydrogen and some ionised metals

Which of the following Hertzsprung-Russell diagram shows the correct location for Star X and Star Y?



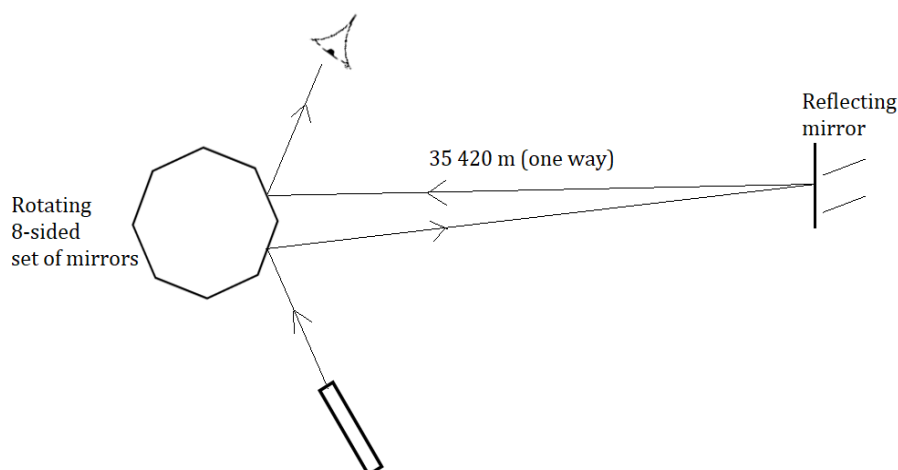
Use the following information for Questions 17 and 18.

The picture shows a setup for Thomson's charge-to-mass ratio experiment.



- What is the purpose of the cathode and anode?
 - (A) To accelerate the charge.
 - (B) To induce a magnetic field to act on the charge.
 - (C) To make the charge travel at a constant velocity
 - (D) To interact with the magnetic field to generate a force.
- Using the electron's kinetic energy, what is the vertical electric field required to produce a valid value for the charge-to-mass ratio?
 - (A) $2.332 \times 10^{-12} \text{ V/m}$
 - (B) $1.876 \times 10^{-6} \text{ V/m}$
 - (C) $1.326 \times 10^3 \text{ V/m}$
 - (D) $1.875 \times 10^3 \text{ V/m}$

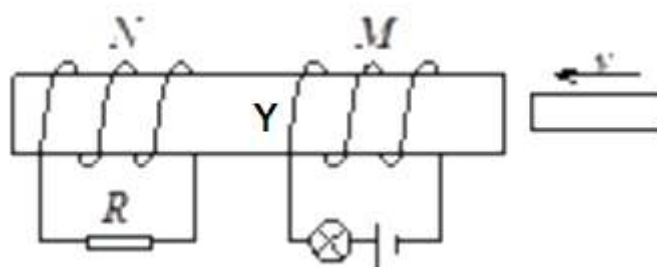
- A scientist used light and rotating mirrors to measure the speed of light. The light was reflected off a mountain 35 420 m away, and the time was determined by using the time it took for the mirror to rotate $\frac{1}{8}$ of a turn, as shown below.



If the speed of light was determined to be 299 796 km/sec, what was the angular velocity (rads^{-1}) of the rotating mirror?

- (A) 3321
- (B) 6648
- (C) 190 280
- (D) 380 559

- A soft iron bar enters the solenoid at a certain velocity and moves to point Y.



Which row correctly identifies the condition of the light in M and current in N?

	<i>The light</i>	<i>The direction of conventional current through the resistor is</i>
(A)	Becomes brighter	Left to right
(B)	Becomes dimmer	Left to right
(C)	Becomes brighter	Right to left
(D)	Becomes dimmer	Right to left



3	7	8		
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Centre Number

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Student Number

2022

Parramatta High School

HIGHER SCHOOL
CERTIFICATE
TRIAL EXAMINATION

Physics

Starting from the left,
please write your student
number NEATLY in the
boxes.

Section I Part A - Multiple Choice

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$

(A) 2

(B) 6

(C) 8

(D) 9

A ☐B ☒C ☐D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒B ☒C ☐D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word **correct** and drawing an arrow as follows.

A ☒B ☒**Correct**C ☐D ☐

Start
Here

1	AO	BO	CO	DO	11	AO	BO	CO	DO
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9 AO BO CO DO

19 AO BO CO DO

10 AO BO CO DO

20 AO BO CO DO

Do NOT write anything, or make any marks below this line



Starting from the left, please write your student number NEATLY in the boxes.

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Centre Number

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Student Number

2022

Parramatta High School

Physics

Section II

80 marks

Attempt Questions 21-34

Allow about 2 hours and 25 minutes for this section

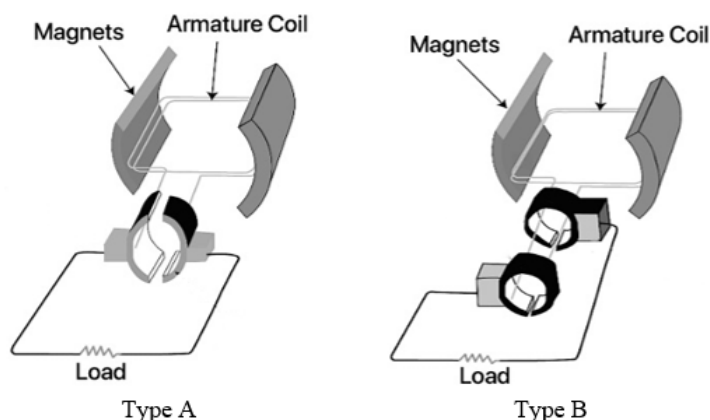
Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.

Show all relevant working out in questions involving calculations.

Extra writing space is provided in an extra writing booklet. If you use this space, clearly indicate which question you are answering.

Question 21 (5 marks)

The following diagram shows two types of generators.



- a) Which of these generators is a DC generator? Justify your answer.

(2 marks)

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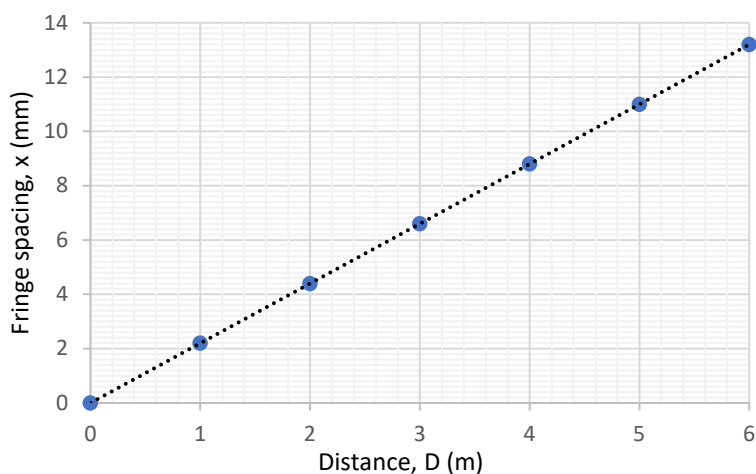
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Question 21 continues on page 2

Question 23 (7 marks)

A group of scientists experimented on diffraction by shining light from a laser with a wavelength of 620 nm on a double slit and observing the interference pattern on a screen situated at a distance D from the slits. The fringe spacing x is measured for a number of different distances, D , and the results are graphed below.



a) Determine the spacing of the slits.

(4 marks)

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b) If the experiment is repeated with the slit spacing doubled, state how the graph and the interference pattern changes.

(3 marks)

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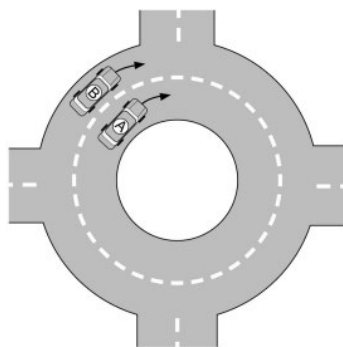
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Question 24 (5 marks)

Cars A and B move around a horizontal dual-lane roundabout at a constant speed of 30 kmh^{-1} , which is the safe speed limit, as shown in the diagram below. (Note: diagram not to scale.)



- a) Compare the acceleration of cars A and B. Include an equation in your answer.

(2 marks)

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- b) Engineers have been asked to redesign the roundabout to increase the safe speed limit while maintaining the same inner and outer radius.

How can the roundabout be redesigned to enable cars to travel safely at a higher speed? Explain your answer.

(3 marks)

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Question 25 (5 marks)

A satellite is orbiting the Earth 4.00×10^3 km above its surface.

Calculate the period of the satellite in hours.

(5 marks)

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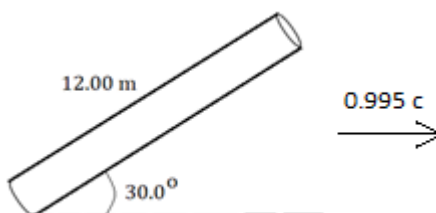
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Question 26 (4 marks)

An observer moving at a speed of $0.995 c$ relative to a rod, as shown in the diagram below, measures its length to be 12.00 m and sees its length to be orientated at 30.0 degrees with respect to its direction of motion.



Determine the proper length of the rod.

(4 marks)

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Question 27 (7 marks)

The model of the atom has changed over time

- a) Explain the experimental evidence that led to Rutherford's model of the atom. (3 marks)

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- b) Explain the importance of the Balmer series in Bohr's explanation for his atomic model. (4 marks)

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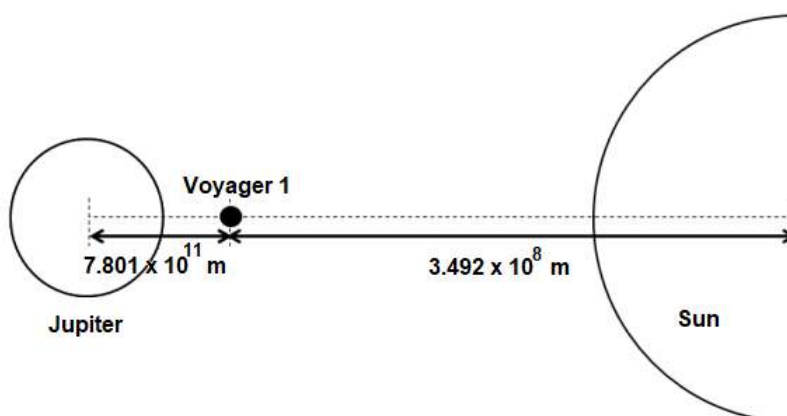
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Question 28 (5 marks)

In 1979, the space probe Voyager 1 was sent to capture images of Jupiter. At the point of closest approach to Jupiter, it was 3.492×10^8 m from the centre of Jupiter and 7.801×10^{11} m from the Sun, as shown in the figure below (not drawn to scale). The mass of Jupiter is 1.898×10^{27} kg, and the mass of the Sun is 1.989×10^{30} kg.



Determine the escape velocity of the Voyager 1 at the point of closest approach to Jupiter (assuming gravitational forces due to other objects are negligible) (5 marks)

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Question 29 (8 marks)

A chain of nuclear reactions in the Sun's core converts four protons into a helium nucleus. The helium nucleus has a mass of 4.00153 amu.

- a) How much energy in MeV is released during the conversion of four protons into a helium nucleus?

(4 marks)

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- b) Explain how nuclear reactions and forces change as main sequence stars evolve to the next life cycle stage.

(4 marks)

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Question 30 (6 marks)

Explain how Chadwick discovered the neutron and derived its mass.

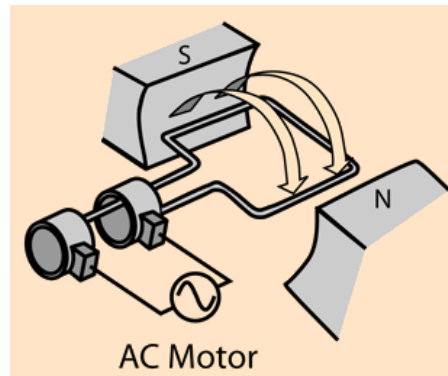
(6 marks)

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

Question 31 (7marks)

Motors are used in everyday life

- a) The picture shows an AC motor.



Explain the relationship between rotation, magnetic flux and back emf as the motor turns. (4 marks)

This image shows a full page of white paper with horizontal dashed lines, typical of primary school writing paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

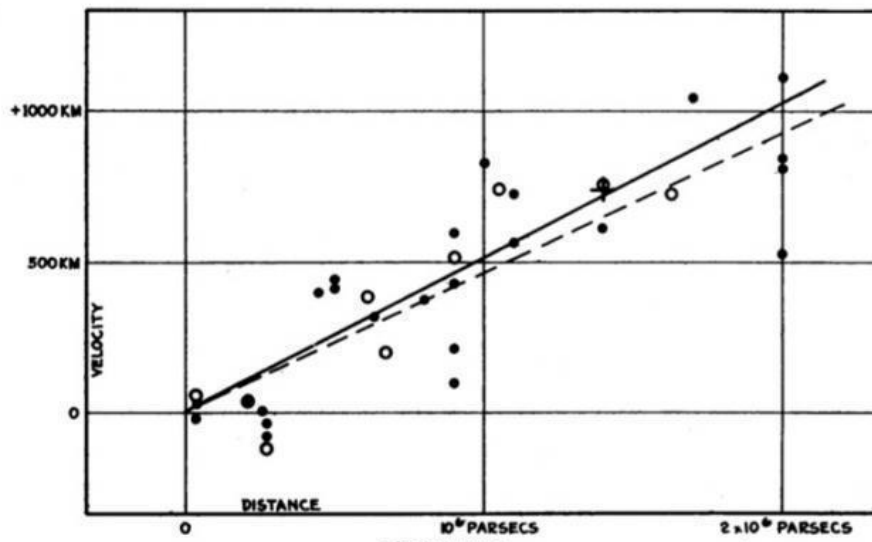
- b) Explain why AC induction motors are widely used in industries. (3 marks)

[illegible]

Question 32 (9 marks)

The following are types of information about space that was collected by astronomers.

Figure 1: Plot of the Velocity-Distance relationship for galaxies.



Velocity-Distance Relation among Extra-Galactic Nebulae.

Figure 2: Spectra of stars

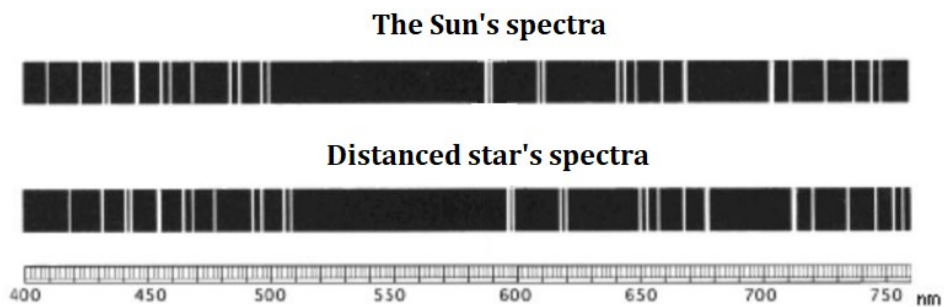
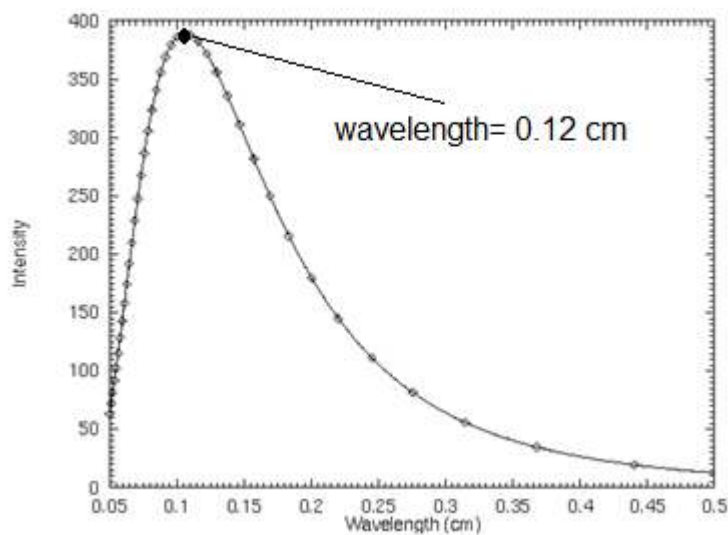


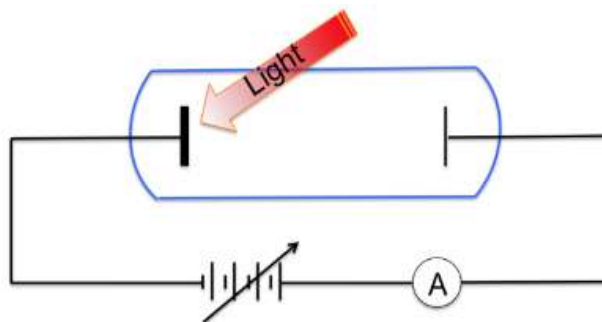
Figure 3: Space thermal radiation measurements from a space probe.



This image is courtesy of Nick Strobel at www.astronomynotes.com

Question 33 (4 marks)

The following shows the setup for the photoelectric effect.



An unknown wavelength of light is shone onto Thorium, and a voltage of 3.56 V is required to stop the emission of photoelectrons. Thorium has a work function of $3.669 \times 10^{-19} \text{ J}$.

The same unknown wavelength of light is then shone onto sodium metal, and photoemission was observed. The threshold frequency of sodium is $6.65 \times 10^{14} \text{ Hz}$.

Determine the velocity of the photoelectrons emitted from the sodium plate. (4 marks)

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Question 34 (8 marks)

Critically analyse how gravitational, magnetic and electric fields affect the motion of a mass in each field with reference to the Laws in Physics. (8 marks)

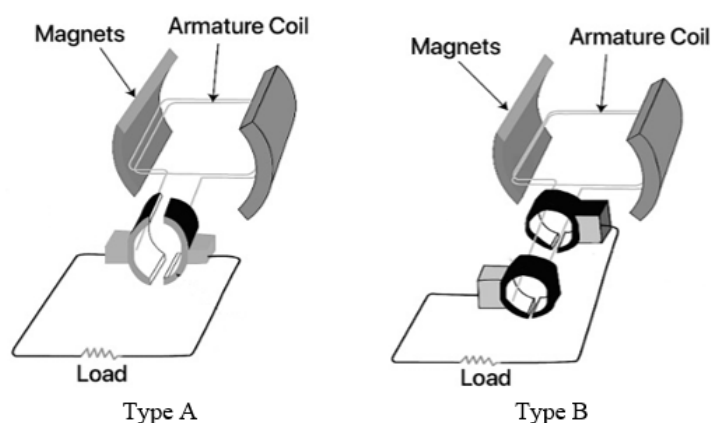
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End of Test

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7	AO	BO	CO	DO	16	AO	BO	CO	DO
8	AO	BO	CO	DO	17	AO	BO	CO	DO
9	AO	BO	CO	DO	18	AO	BO	CO	DO
10	AO	BO	CO	DO	19	AO	BO	CO	DO

Question 21 (5 marks)

The following diagram shows two types of generators.



- a) Which of these generators is a DC generator? Justify your answer. (2 marks)

Marking Criteria	Marks
• Provide explanation that links to the function of split ring commutator	2
• Identify that type A is a DC generator OR provide some relevant information	1

Type A is a DC generator. Because it has a split ring commutator which rectifies the current. As the coil rotates, the induced emf of the coil is AC but the brushes reverse the side of the coil causing the supplied emf to the external circuit to be in one direction only, hence DC.

- b) Evaluate the effectiveness of each type of generator in large-scale power transmission. (3 marks)

Marking Criteria	Marks
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Thoroughly explain the efficiency in large-scale power transmission for BOTH generators	3
Briefly explain the efficiency in large-scale power transmission for BOTH generators OR Thoroughly explain the efficiency in large-scale power transmission for ONE generator	2
Provides some relevant information	1

Type B is better since AC generators provide alternating current hence can be used in step-up transformers to produce high voltage needed for transmission. This results in more efficient transmission of electricity between power station and households as only small currents are required. The smaller the currents, the less energy is lost due to heating effects caused by the resistance of wires to the flow of electric current. However DC generators provide direct current hence it can't be used with transformer thus the energy lost on transmission lines will be much higher.

Question 22 (4 marks)

How have different theories in Physics changed the definition of SI units?

(4 mark)

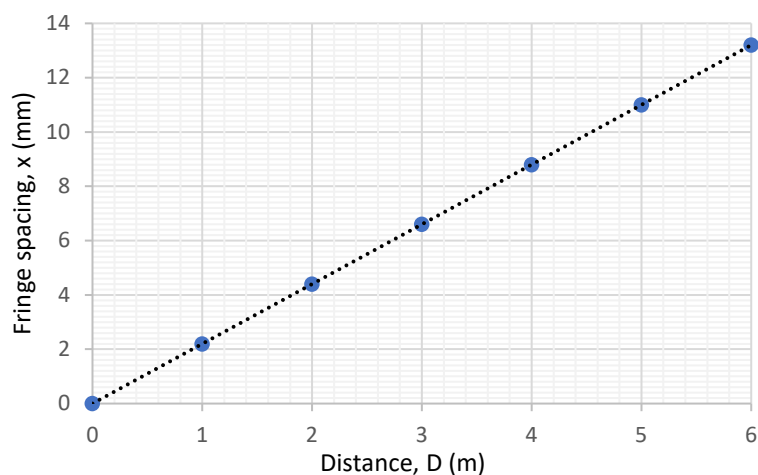
Marking Criteria	Marks
Relates interaction b/w parallel current carrying wires to ampere and relativity to the metre and time	4
Describes amperes with the interaction b/w parallel current carrying wires to ampere and the metre and time to relativity	3
Describes amperes with the interaction b/w parallel current carrying wires to ampere OR the metre and time to relativity	2
Provides some relevant information	1

Sample answer

Previously the ampere was defined as the current that would deposit precisely 0.001118 grams of silver per second from a silver nitrate solution. However, with Oersted's discovery of a current carrying conductor producing a magnetic field and the interactions between two current conductors, changed the definition of ampere as the constant current which, if maintained in two straight parallel conductors of infinite length of negligible circular cross-section and placed one metre apart in a vacuum, would produce between these conductors a force equal to 2×10^{-7} . However, with special relativity the constant speed of light has changed our definition for time is no longer the transition of cesium-133 but now is the switch to atomic time, as a result the meter was redefined, so its definition depends on the speed of light, and the definition of a second—it's the distance light travels in 1299792458 seconds. This led to the most current definition of the ampere is by taking the fixed numerical value of the elementary charge e to be $1.602\,176\,634 \times 10^{-19}$ when expressed in the unit C, which is equal to A s, where the second is defined in terms of $\Delta\nu$.

Question 23 (7 marks)

A group of scientists experimented on diffraction by shining light from a laser with a wavelength of 620 nm on a double slit and observing the interference pattern on a screen situated at a distance D from the slits. The fringe spacing x is measured for a number of different distances, D , and the results are graphed below.



c) Determine the spacing of the slits.

(4 marks)

Marking Criteria	Marks
• Correctly calculate the gradient to determine the spacing correctly	4
• Correctly calculate the gradient and/or derive the formula to determine the spacing correctly	3
• Correctly calculate the gradient or derive the formula to determine the spacing correctly	2
• Provides some relevant information	1

Sample answer:

$$d \sin \theta = m\lambda \text{ and } \tan \theta = \frac{x}{D} = \text{gradient of graph}, \text{ so Gradient} = \frac{11 \times 10^{-3} - 0}{5 - 0} = \frac{11 \times 10^{-3}}{5}$$

At small angles, $\sin \theta = \tan \theta$

$$d \times \text{gradient} = 1 \times 620 \times 10^{-9}$$

$$d = \frac{1 \times 620 \times 10^{-9}}{\frac{11 \times 10^{-3}}{5}} = 2.82 \times 10^{-4} \text{ m}$$

d) If the experiment is repeated with the slit spacing doubled, state how the graph and the interference pattern changes.

(3 marks)

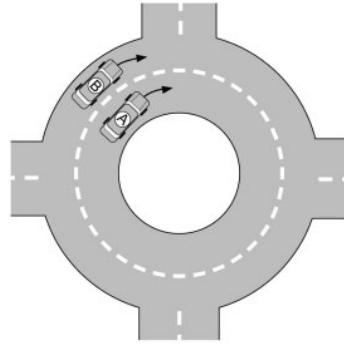
Marking Criteria	Marks
• Describes how the gradient of the graph and interference pattern changes with double spacing	3
• Describes how the gradient of the graph and/or interference pattern changes with double spacing	2
• Provides some relevant information	1

Sample answer

If the slit spacing was double then the fringe width spacing will decrease meaning less interference. So the graph will be shallower with a gradient of 1.1.

Question 24 (5 marks)

Cars A and B move around a horizontal dual-lane roundabout at a constant speed of 30 kmh^{-1} which is the safe speed limit, as shown in the diagram below. (Note: diagram not to scale.)



c) Compare the acceleration of cars A and B. Include an equation in your answer.

(2 marks)

$a_c = \frac{v^2}{r}$	1
as $r_A < r_B$ and v is constant for both vehicles and the same speed then $a_c \propto \frac{1}{r}$	1

therefore $a_c A > a_c B$ since $r_A < r_B$

d) Engineers have been asked to redesign the roundabout to increase the safe speed limit while maintaining the same inner and outer radius.

How can the roundabout be redesigned to enable cars to travel safely at a higher speed? Explain your answer.

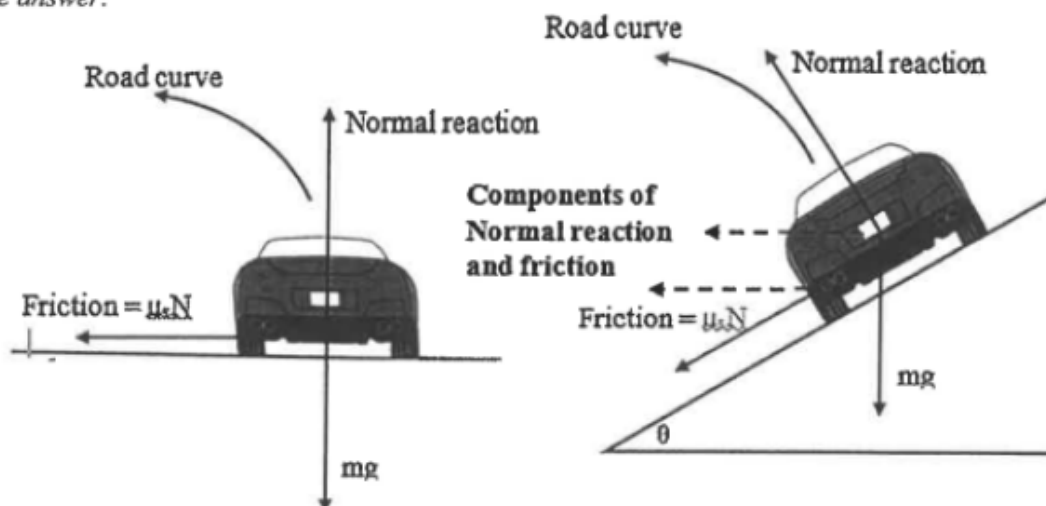
(3 marks)



Description	Marks
the roundabout will need to be banked/inclined plane	1
this means that the normal force can provide a horizontal component inwards which contributes to the centripetal force required	1
thus the friction force needed can stay within safe limits. (On a horizontal track, friction is the only force providing centripetal force.)	1
or	
make the surface rougher	1
this will increase friction	1
thus the friction force is larger and can stay within safe limits. (On a horizontal track, friction is the only force providing centripetal force.)	1
Total	3

Criteria	Marks
• Identifies ALL forces acting, draws components of forces on diagram and compares the magnitudes of their contribution to centripetal force	4
• Identifies all forces acting and draws components of forces on diagram contributing to centripetal force	3
• Identifies the contribution of the normal reaction component to centripetal force	2
• Provides some relevant information	1

Sample answer:



On the horizontal road, the friction alone provides the centripetal force to hold car in curve. On the banked road, only a component of the frictional force is acting horizontally but the horizontal component of the normal reaction force contributes. This is significantly larger than the reduction in the frictional force towards the centre of the motion. Therefore, the net centripetal force is larger and since mass and radius of curvature are the same, the car can go faster around the banked curve.

Question 25 (5 marks)

A satellite is orbiting the Earth 4.00×10^3 km above its surface.

Calculate the period of the satellite in hours.

(5 marks)

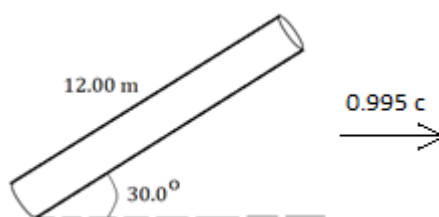
(a) Calculate the period of the satellite.

(5 marks)

Element	Description	Marks
converts km to m	$T^2 = 4\pi^2 (4.0 \times 10^6 + 6.37 \times 10^6)^3 / (6.67 \times 10^{-11} \times 5.97 \times 10^{24})$	1
adds altitude to radius of Earth		1
cubes distance and square roots answer		1
correct answer	1.05×10^4 s	1
converts to hours	2.92 hours	1
Total		5

Question 26 (4 marks)

An observer moving at a speed of $0.995c$ relative to a rod, as shown in the diagram below, measures its length to be 12.00 m and sees its length to be orientated at 30.0 degrees with respect to its direction of motion.



Determine the proper length of the rod.

(4 marks)

Marking Criteria	Marks
• Provides steps to correctly calculate the proper length using correct contracted length	4
• Provides steps to correctly calculate the proper length	3
• Provides steps to calculate the proper length	2
• Provides some relevant information	1

Sample answer

Length undergoing contraction

$$l = 12 \cos 30$$

$$l_o = 12 \cos 30 \times \sqrt{1 - 0.995^2}$$

$$\text{Proper length} = \sqrt{(12 \cos 30 \times \sqrt{1 - 0.995^2})^2 + 12^2 \sin^2 30} = 104.23\text{m}$$

Question 27 (7 marks)

The model of the atom has changed over time

c) Explain the experimental evidence that led to Rutherford's model of the atom.

(3 marks)

Marking Criteria	Marks
• Relates experimental evidence to features of the Rutherfords model	3
• Outlines experimental evidence that led to the Rutherfords model	2
• Provides some relevant information	1

Sample answer:

Geiger-Marsden fired alpha particles which were positively charged at a gold foil and deflection was measured by a florescent screen, they found that the majority of the alpha particles traveled straight through, however, some were deflected indicating some positive charge in the centre and there were some which were nearly deflected straight back which lead to Rutherford concluding that the atom was mostly empty space (majority passing through) with a dense positive nucleus that caused the sharp deflection of the alpha particles.

d) Explain the importance of the Balmer series in Bohr's explanation for his atomic model.

(4 marks)

Marking Criteria	Marks
• Provides a thorough understanding of Bohr's use of Balmer's emission series in his model of the atom	4

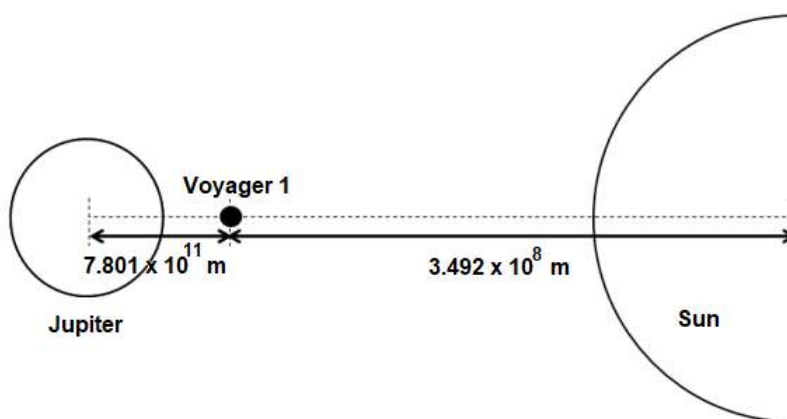
• Provides a sound understanding of Bohr's use of Balmer's emission series in his model of the atom	3
• Provides some understanding of Bohr's model of the atom or Balmer's emission series	2
• Provides some relevant information	1

Sample answer:

The Balmer series is vital to Bohr's model as he applied Planck's discrete packet quanta of energy to describe the energy emission of the hydrogen spectrum in the visible light region. He proposed that Balmer emission was due to electrons jumping from an excited (higher energy shell) to a lower energy shell, they only emit or absorb a specific quanta of energy when they move from one energy level to another- which was his second postulate that electron in stable orbits does not emit electromagnetic radiation. The Balmer-Rydberg equation that can theoretically calculate hydrogen spectra line further support Bohr's application of quanta to his atomic theory.

Question 28 (5 marks)

In 1979, the space probe Voyager 1 was sent to capture images of Jupiter. At the point of closest approach to Jupiter, it was 3.492×10^8 m from the centre of Jupiter and 7.801×10^{11} m from the Sun, as shown in the figure below (not drawn to scale). The mass of Jupiter is 1.898×10^{27} kg, and the mass of the Sun is 1.989×10^{30} kg.



Determine the escape velocity of the Voyager 1 at the point of closest approach to Jupiter (assuming gravitational forces due to other objects are negligible) (5 marks)

Marking Criteria	Marks
• Correctly calculate escape velocity with correct sig fig	5
• Correctly calculate escape velocity	4
• Provides steps to correctly calculate the proper length	3
• Provides steps to calculate the proper length	2
• Provides some relevant information	1

Sample answer

Resultant potential at the point

$$U = U_s + U_J$$

$$U = m_s \left(-\frac{GM_s}{r_s} \right) + \left(-\frac{GM_J}{r_J} \right)$$

$$U = m_s \left(-\frac{6.67 \times 10^{-11} \times 1.989 \times 10^{30}}{7.801 \times 10^{11}} \right) + \left(-\frac{6.67 \times 10^{-11} \times 1.898 \times 10^{27}}{3.492 \times 10^{11}} \right)$$

Escape velocity

$$\frac{1}{2} m_s v_{esc}^2 = m_s \left(-\frac{6.67 \times 10^{-11} \times 1.989 \times 10^{30}}{7.801 \times 10^{11}} \right) + \left(-\frac{6.67 \times 10^{-11} \times 1.898 \times 10^{27}}{3.492 \times 10^{11}} \right)$$

$$v_{esc} = \sqrt{2 \left| \left(-\frac{6.67 \times 10^{-11} \times 1.989 \times 10^{30}}{7.801 \times 10^{11}} \right) + \left(-\frac{6.67 \times 10^{-11} \times 1.898 \times 10^{27}}{3.492 \times 10^{11}} \right) \right|}$$

$$= 18422.847 \text{ ms}^{-1} = 1.842 \times 10^4 \text{ ms}^{-1}$$

Question 29 (8 marks)

A chain of nuclear reactions in the Sun's core converts four protons into a helium nucleus. The helium nucleus has a mass of 4.00153 amu.

- c) How much energy in MeV is released during the conversion of four protons into a helium nucleus?

(4 marks)

Marking Criteria	Marks
• Correctly calculate the energy in MeV	4
• Correctly calculate the energy in Joules	3
• Provides steps to calculate the energy	2
• Provides some relevant information	1

Sample answer



$$\text{Mass defect} = 4(1.673 \times 10^{-27}) - (4.00153 \times 1.661 \times 10^{-27}) = 4.545867 \times 10^{-29}$$

$$\text{Energy} = 4.545867 \times 10^{-29} \times (3.00 \times 10^8)^2 = 4.0912803 \times 10^{-12} \text{ J} = \frac{4.0912803 \times 10^{-12}}{1.602 \times 10^{-19}} \times 10^{-6} = 25.54 \text{ MeV or } 25.49$$

- d) Explain how nuclear reactions and forces change as main sequence stars evolve to the next life cycle stage.

(4 marks)

Marking Criteria	Marks
• Provides a thorough understanding of the balance between nuclear reaction pressure and gravity in main sequence stars and red giant stars	4
• Provides a sound understanding of the balance between nuclear reaction pressure and gravity in main sequence stars and/or red giant stars	3
• Provides some understanding of nuclear reaction pressure and gravity in main sequence stars and/or red giant stars	2

Provides some relevant information	1
--	---

Sample answer

In a main sequence star, the star can either undergo proton-proton nuclear fusion or CNO fusion. The outwards pressure particle due to the fusion process and the release of radiation balances the inward gravitational force. Small stars undergo hydrogen-burning core as it moves to the red giant stage the core is predominately helium, and the outward pressure due to is hydrogen burning in the shell is greater than the gravitational force causing the star to expand eventually there will be less hydrogen burning and more helium in the core which increased the gravitational force causing the star to collapse and shrink in size. Bigger stars burning CNO undergoes a similar force mechanism however, they will burn their fuel faster and collapse faster either into a neutron star or a black hole depending on their mass.

Question 30 (6 marks)

Explain how Chadwick discovered the neutron and derived its mass.

(6 marks)

Marking Criteria	Marks
- Explain the experiment and the results that led to Chadwick's discovery of the neutron - Explain and shows mathematically how Chadwick used the conservation of energy and momentum in deriving a mass for the neutrons	6
- Explain the experiment and the results that led to Chadwick's discovery of the neutron - Explain how Chadwick used the conservation of energy and momentum in deriving a mass for the neutrons	5
- Explain the experiment and the results that led to Chadwick's discovery of the neutron - Explain how Chadwick used the conservation of energy and momentum in deriving a mass for the neutrons	4
Outline the experiment and the result that led to Chadwick's discovery of the neutron	3
And/or Outlines Chadwick used the conservation of energy and momentum in deriving a mass for the neutrons	
OR - Outline the experiment and the result that led to Chadwick's discovery of the neutron - Outlines Chadwick used the conservation of energy and momentum in deriving a mass for the neutrons	2
Provides some relevant information	1

Sample answer

Chadwick used the Curies and Bothe and Becker's experiment where they fired alpha particles at Beryllium and found a penetrating radiation that can then knock out protons from paraffin wax or even nitrogen, since it's the same particle that can knock out two different mass atoms and applying the conservation of energy, he concluded that it is not gamma radiation but a neutral particle that is undergoing elastic collision with the hydrogen and nitrogen. Measuring the recoil velocity and apply the conservation of momentum the mass of the neutron was calculated.

A neutron mass m_n and velocity v_n has a head on collision with the stationary hydrogen nucleus of mass m_h . Let the velocity of the neutron after the collision be v_{nf} , and the maximum velocity of the recoiling hydrogen nucleus of v_h .

By applying conservation of energy and momentum he derives the following equation

$$\frac{1}{2}m_n v_n^2 = \frac{1}{2}m_n v_{nf}^2 + \frac{1}{2}m_h v_h^2 \text{ and } m_n v_n = m_n v_{nf} + m_h v_h$$

Equation v_{nf}

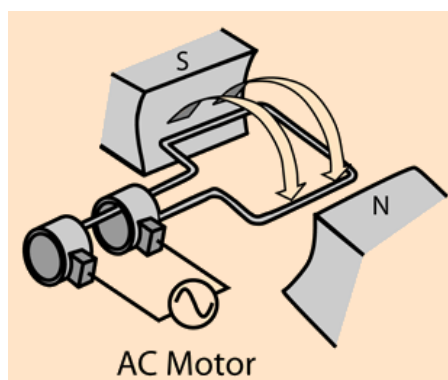
$v_h = \frac{2m_n}{m_n + m_H} v$ using the recoil velocity for hydrogen and nitrogen (m_N) the mass of the neutron can be determined.

That is $\frac{V_H}{V_N} = \frac{m_n + m_N}{m_n + m_H}$

Question 31 (7marks)

Motors are used in everyday life

- c) The picture shows an AC motor.



- a) Explain the relationship between rotation, magnetic flux and back emf as the motor turns. (4 marks)

Marking Criteria	Marks
Explain the relationship between rotation (angular velocity), magnetic flux (sine wave) and back emf (change in flux induces back emf due to LOCE)	4
- Describe the relationship between rotation, magnetic flux and back emf - OR - Explain the relationship between magnetic flux and back emf as the motor turns but missing the rotational speed	3
Outline the relationship between rotation and magnetic flux OR back emf	2
Provides some relevant information	1

As the AC current passes through the coil, a torque is generated on the coil to allow the motor to run at the frequency of the sine wave. When the motor starts to rotate, the coil experiences a change in magnetic flux which also changes at the frequency of the sine wave but out of phase. This change in magnetic flux, according to Faraday's Law, will induce a back EMF that is in the opposite direction of the supplied emf, based on Lenz's Law and the Law of conservation.

ω it turns
 $\downarrow$
 $A = \pi r^2 \cdot \frac{\omega}{2\pi} = \pi r^2 \cdot \frac{\omega t}{2\pi}$
 $= \frac{1}{2} r^2 \omega t$
 $\Delta \phi = \Delta B A$ magnetic flux
 $\downarrow$
 $E = -\frac{\Delta \phi}{\Delta t}$
 $= -\frac{\Delta B \cdot \frac{1}{2} r^2 \omega t}{\Delta t}$
 $= -\frac{1}{2} B \cdot r^2 \omega$ change in flux
 $\hookrightarrow$ back emf $\leftarrow$
 opposite direction

b) Explain why AC induction motors have replaced some DC motors.

(3 marks)

Marking Criteria	Marks
Explain at least two features and characteristics of AC induction motor that are better than DC motors	3
Describe at least two features and characteristics of AC induction motor that are better than DC motors	2
Provides some relevant information	1

AC induction motors are more widely used due to the following reasons:

Three-phase AC induction motors are **self-starting** by rotating the magnetic field hence change the magnetic flux and create a torque. Whereas simple AC motor requires some external equipment to start the motor.

The speed of AC induction motors can be **controlled** easily but the AC motor cannot change its speed. Many industrial applications require different speed for different loads (torque).

AC induction motors, compared to simple AC motor and the DC motor, do not have brushes or slip rings. This eliminates the friction factor and the sparks hence it's very **safe**. Also its lifespan is longer and requires less maintenance.

AC induction motors do not require permanent magnets. It relies on the change of currents in the stator coils. Therefore it's less costly and complex. The size of the motor is also significantly smaller than DC motors.

AC induction motors can be used along with transformers to generate electricity for large scale electricity transmission.

Question 32 (9 marks)

The following are types of information about space that astronomers collected.

Figure 1: Plot of the Velocity-Distance relationship for galaxies.

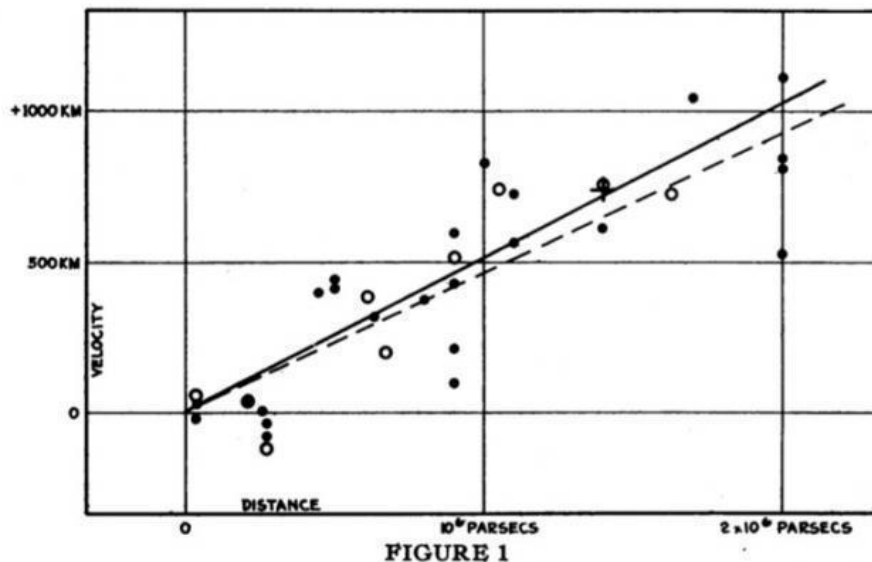


FIGURE 1
Velocity-Distance Relation among Extra-Galactic Nebulae.

Figure 2: Spectra of stars

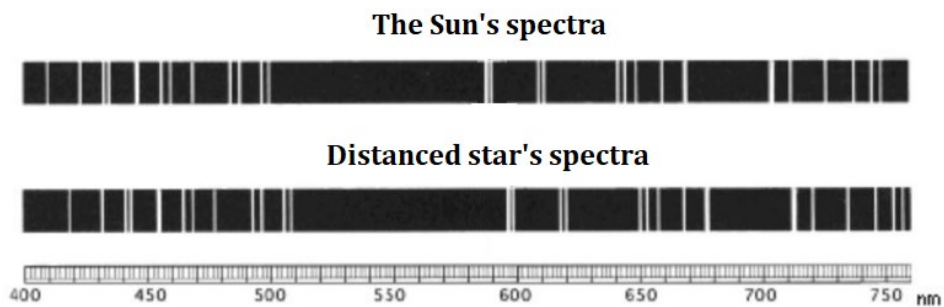
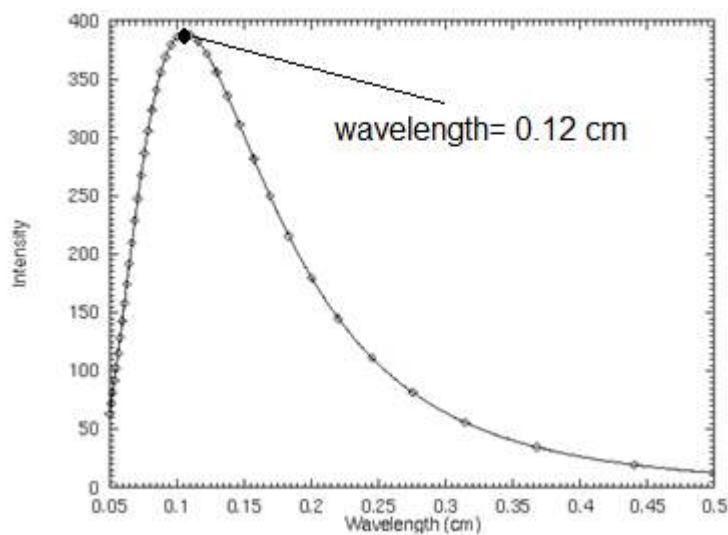


Figure 3: Space thermal radiation measurements from a space probe.



This image is courtesy of Nick Strobel at www.astronomynotes.com

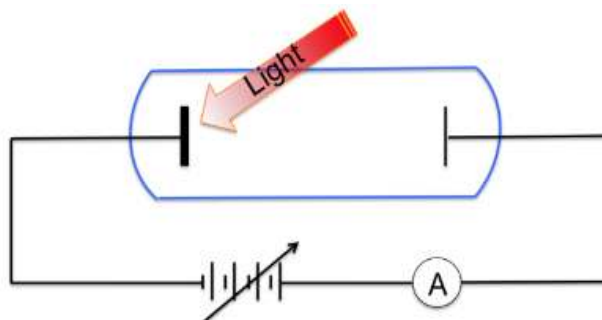
Analyse how these three pieces of information are used to support the Big Bang Theory.

(9 marks)

Marking Criteria	Marks
- Reference the emission spectrum of distance stars to the sun spectra and relate it to the doppler effect showing that the stars are exhibiting red shift with the increase in wavelength and are moving away as evidence of the Big Bang Theory-expansions from a single point - Thoroughly explain the feature in figure 1, linking it to Hubble's law expansion of space and relate it to emission spectra as evidence for the expanding universe, which supports the Big Bang Theory - Explain the use of blackbody radiation curve when measuring the temperature of the universe to calculate the wavelength of the cosmic background radiation to be in the microwave range which are the remanent of the energy released from the singularity point- due to conversion to matter as support for the big bang theory	9
- Reference the emission spectrum of distance stars and relate it to the doppler effect showing that the stars are moving away as evidence of the Big Bang Theory - Explain the feature in figure 1, linking it to Hubble's law and the expanding universe, which supports the Big Bang Theory - Explain the use the blackbody radiation curve to calculate the wavelength of the cosmic background radiation and/or explain how it provides evidence for the Big Bang Theory	7-8
- Explain the feature in figure 1, linking it to Hubble's law and/or the expanding universe, which supports the Big Bang Theory AND/OR - Reference the emission spectrum of distance stars and relate it to the doppler effect showing that the stars are moving away as evidence of the Big Bang Theory AND/OR - Explain how background radiation provides evidence for the Big Bang Theory	5-6
- Outline the feature in figure 1, linking it to Hubble's law, which supports the Big Bang Theory AND/OR - Reference the emission spectrum to the doppler effect showing that the stars are moving away as evidence of the Big Bang Theory AND/OR - Explain cosmic background radiation and how it provides evidence for the Big Bang Theory	3-4
Provides some relevant information	1-2

Question 33 (4 marks)

The following shows the setup for the photoelectric effect.



An unknown wavelength of light is shone onto Thorium, and a voltage of 3.56 V is required to stop the emission of photoelectrons. Thorium has a work function of 3.669×10^{-19} J.

The same unknown wavelength of light is then shone onto sodium metal, and photoemission was observed. The threshold frequency of sodium is 6.65×10^{14} Hz.

Determine the velocity of the photoelectrons emitted from the sodium plate. (4 marks)

Marking Criteria	Marks
• Correctly calculate the velocity of the photoelectrons	4
• Calculate the velocity of the photoelectrons	3
• Provides steps to calculate the energy	2
• Provides some relevant information	1

Sample answer

$$\phi = hf - qVo,$$

$$3.669 \times 10^{-19} = 6.626 \times 10^{-34} f - 1.602 \times 10^{-19} \times 3.56$$

$$\text{So } f = \frac{3.669 \times 10^{-19} + 1.602 \times 10^{-19} \times 3.56}{6.626 \times 10^{-34}} = 1.414382735 \times 10^{15}$$

$$K = hf - hf_0 = h(1.414382735 \times 10^{15} - 6.65 \times 10^{14}) = 4.96541 \times 10^{-19}$$

$$\frac{1}{2} \times 9.109 \times 10^{-31} \times v^2 = 4.96541 \times 10^{-19}, \text{ so } v = \sqrt{\frac{4.96541 \times 10^{-19}}{\frac{1}{2} \times 9.109 \times 10^{-31}}} = 1.04 \times 10^6 \text{ m/s}$$

Question 34 (8 marks)

Critically analyse how gravitational, magnetic and electric fields affect the motion of a mass in each field with reference to the Laws in Physics. (8 marks)

Marking Criteria	Marks
• Demonstrate understanding of Law of conservation of Energy AND • Demonstrate extensive understanding of the force and motion of a mass in each field, including the correct Laws of Physics	8
• Demonstrate understanding of Law of conservation of Energy AND/OR • Demonstrate some thorough understanding of the force and motion of a mass in each field, including the correct Laws of Physics	6-7
• Demonstrate understanding of Law of conservation of Energy AND/OR • Demonstrate sound understanding of the force and motion of a mass in each field, including the correct Laws of Physics OR • Demonstrate good understanding of the force and motion of a mass in two fields, including the correct Laws of Physics	4-5
• Demonstrate understanding of Law of conservation of Energy AND/OR • Demonstrate some basic understanding of the motion in each field	2-3
• Provides some relevant information	1

Sample answer:

Gravitational field is exerted by all objects with a mass. Gravitational field is treated as uniform near Earth's surface. It attracts objects towards the center of the Earth. Depends on whether the object has an initial horizontal velocity, the motion could either be parabolic or vertical. However, satellites, planets and other celestial objects experience non-uniform gravitational fields. According to Newton Law of Universal Gravitation, the gravitational force is $F = GMm/r^2$. For planets to orbit the Sun, they must have a tangential orbital velocity $v = \sqrt{GM/r}$ and gravity will provide the centripetal force. The radius of the elliptical motion can be determined by Kepler's 2nd and 3rd Law. For satellites to orbit the Earth, it also required an orbital velocity. However, satellites move in a free falling motion and this leads to the loss of energy. Extra kinetic energy is required to transform to potential energy to maintain their orbits.

Electric field exerts force on all charged masses. Neutral mass will not be affected by electric field. The force on the charged mass is given by $F = qE = qV/d$ in uniform E field between two plates or $F = kq_1q_2/r^2$ in non-uniform E field between two charges. A positive charge will accelerate at $a = qE/m$ in the direction of the electric field and a negative charge will accelerate against the direction of the field. However, in both scenarios, the charged mass will gain kinetic energy due to the change in electrical potential energy.

If the charged particle is initially at rest, the force on the particle will cause it to accelerate towards the positive/negative plate based on the charge. If the charged particle is moving, depends on the direction of the initial velocity, the motion can be parabolic or parallel/perpendicular to the E field.

Magnetic field exerts a force on moving charged particles. The force is always perpendicular to the velocity of the charge hence it is a centripetal force and causes the charged particle to follow either a circular motion (initial velocity is perpendicular to the B field) or a helical motion (initial velocity enters B field at an angle). The force is given by $F = qvB\sin\theta$. Like E field, magnetic force for positive and negative charged moving particles are opposite. Any conductive material or ferromagnetic material in B field will receive magnetic force as well. *may talk about wires and motors as well...

*should talk about correct Laws such as Law of Conservation of Energy, Newton's Laws of Motion, Coulomb's Law. Not necessary Faraday or Lenz's unless you talked about the motor effect.